

SYSTEM SURVEILLANCE USING KEYLOGGER

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Abstract— In the contemporary business landscape, safeguarding data and ensuring its recovery in unforeseen circumstances is vital. To address this need, keyloggers, also known as keyboard capturing tools, have gained prominence. These tools covertly record keystrokes, providing a means to monitor user actions without their awareness. The novelty of this project lies in its easy implementation, requiring minimal investment, making it accessible to any organization. Initial results show promising outcomes in data tracking, bolstering both security and retrieval processes.

Keywords: *Keylogger, System Surveillance, Data Security, Data Recovery, Data Safeguarding*

I. INTRODUCTION

In today's corporate environment, the significance of data security and recovery cannot be overstated. Various unforeseen events, such as power outages, can lead to the loss or corruption of crucial working files. To address these challenges, keyloggers, commonly referred to as keyboard capturing tools, have emerged as a reliable solution. Keyboard capturing involves the discreet recording of keystrokes, allowing for the covert monitoring of user actions. This keylogger project takes a proactive approach by capturing every keystroke made by a user and promptly sending this information to an administrator's email address. The application operates silently, ensuring that users are unaware of the ongoing monitoring process. By utilizing log files, the keylogger facilitates data recovery in situations where working files are compromised. Additionally, the tool proves beneficial for recalling forgotten emails or URLs. The main objective of this application is to enhance data security, confidentiality, and recovery for IT infrastructures. By providing a seamless and covert means of tracking and logging keystrokes, the keylogger contributes to a robust defense against data loss and unauthorized access.

The paper begins with a comprehensive review of existing literature on data security, recovery methods, and the role of keyloggers in addressing these concerns. This section establishes the context and significance of the project within the broader research landscape. The methodology section outlines the technical approach taken in developing the keylogger application. It describes the software architecture, data collection methods, and email transmission protocols employed

to achieve the desired functionality. The former is followed by the results section, which presents the findings of the keylogger application's performance in real-world scenarios. It includes metrics such as keystroke capture accuracy, email transmission efficiency, and user feedback. The discussion interprets these results in the context of the project objectives, highlighting strengths, limitations, and potential areas for improvement.

By comprehensively tracking every keystroke and transmitting this information to the administrator via email within a predefined time frame, the application successfully enhances confidentiality and facilitates data recovery for IT infrastructures. Operating seamlessly in the background, it serves as a valuable asset for organizations seeking to bolster their data security measures and ensure the retrieval of critical information in adverse situations.

II. LITERATURE SURVEY

The continuous analysis of digital systems across various software environments using keyloggers, as highlighted by Bhavitha et al. [1], underscores the pressing need for heightened security measures. Their study evaluates different system softwares' security postures through a comparative analysis, identifying vulnerabilities and advocating for robust security measures to safeguard user data integrity and confidentiality. Meanwhile, Kazi et al.'s research focuses on proposing and evaluating methodologies to effectively detect keyloggers [2], aiming to fortify digital systems against their pervasive threat. By utilizing machine learning and heuristic-based approaches, they strive to bolster security mechanisms and foster user trust in digital environments. Additionally, Sagioglu & Canbek shed light on the increasing threats posed by keyloggers to computer security and privacy [3]. Their study provides a comprehensive understanding of keyloggers, exploring their evolving techniques, threat extent, and potential countermeasures. They emphasize the critical need for proactive measures to mitigate keylogger risks, contributing valuable insights to ongoing discussions on strengthening computer security protocols and safeguarding user privacy in the digital realm.

The collective research endeavors highlight various aspects of keyloggers and their implications for cybersecurity. Prajapati et al. conduct a comprehensive analysis, elucidating keylogger functionalities, potential threats, and countermeasures to fortify cybersecurity defenses [4]. Their study categorizes keyloggers, examines their functionalities, and proposes detection and mitigation strategies to inform cybersecurity professionals in

developing robust defense strategies. Rai et al. delve into the intricate relationship between encryption and keylogging technologies, exploring their synergies, conflicts, and implications for computer system security [5]. Through a systematic review, they evaluate the effectiveness of encryption and keylogging technologies, offering insights for developing comprehensive security strategies against evolving cyber threats. Wajahat et al. address the detection and prevention of userspace keyloggers by correlating I/O activities with simulated user input [6]. Their research introduces a robust detection method for unprivileged userspace keyloggers, enhancing security against covert keystroke recording without false positives or negatives. Collectively, these studies contribute valuable insights into understanding keyloggers' functionalities, the relationship between encryption and keylogging, and effective detection methods to bolster cybersecurity defenses.

Singh et al. review keylogger programs, types, characteristics, and methodologies, highlighting limitations of traditional antivirus software against known keyloggers and proposing detection and prevention methods, including anti-malware software and vigilance with browser-related keyloggers [7]. Adhikary et al. propose a model utilizing fabricated passwords on untrusted machines to deceive keylogging and video capturing software, enhancing protection against data theft [8]. Dadkhah et al. introduce undetectable keyloggers coded in C#, testing against popular antivirus programs to highlight potential failures of existing techniques against advanced variants [9]. Bhardwaj & Goundar address keylogger trojans' evolving threat, proposing an updated taxonomy and illustrating a potent keylogger technique that evades detection, emphasizing the need for robust cybersecurity measures [10]. Wazid et al. combine honeypot technology with malware detection to fortify defenses against keylogger spyware attacks [11], while Kavya & Suganya conduct a comprehensive literature review on keyloggers, password attacks, and mitigation techniques [12]. Santoki develops a keylogger detector employing a signature database and network monitoring to detect known keyloggers, providing insights into keylogger behavior but with limitations regarding manual execution and detection of new variants [13]. Bhujel & Priya introduce a software keylogger developed using Python and the pynput library, clandestinely capturing keystrokes, mouse clicks, and window titles without the user's knowledge, raising ethical concerns about privacy violation and potential malicious use [14]. Ingle et al. present "Key Log Spy," a Python-based software keylogger designed to surreptitiously capture keystrokes, system information, network details, clipboard data, and screenshots without user knowledge, falling short in addressing crucial ethical and legal concerns [15]. Vinothkumar addresses the rise of keyloggers and mobile malware, emphasizing the need for anomaly detection, machine learning algorithms, signature-based detection, and behavioral analysis to mitigate cybersecurity threats [16]. Ahsan Wajahat et al. propose a strategy for detecting user space keyloggers, employing detection manner techniques to match I/O of all processes with simulated user activity for effective detection and prevention [17]. Ayo et al. introduce CaFISKLD, a combinatorial-based fuzzy inference system for keylogger detection, aiming to improve classification accuracy and pattern discovery [18]. Kazi et al. conduct a comprehensive review of

existing literature on keylogger detection methodologies, categorizing techniques and proposing future directions for research [19]. Samaria & Adil introduce a novel hybrid approach for keylogger detection, integrating signature-based detection, anomaly detection, and behavior-based analysis to enhance accuracy and robustness [20].

III. PROPOSED METHODOLOGY

The proposed methodology of the keylogger software involves the development of a program designed to monitor and record keyboard input on macOS systems. The program utilizes the Core Graphics (CG) framework to establish an event tap, enabling it to intercept keyboard events.

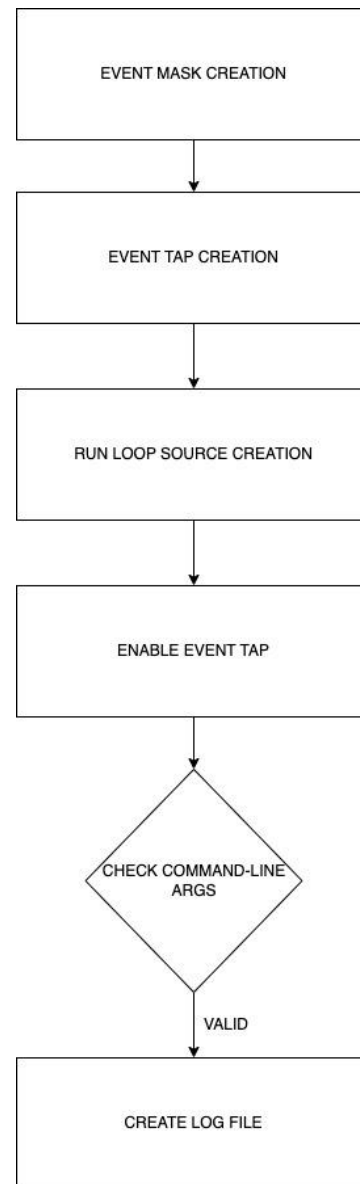


Fig1. Core Graphics (CG) Event Tapping Architecture

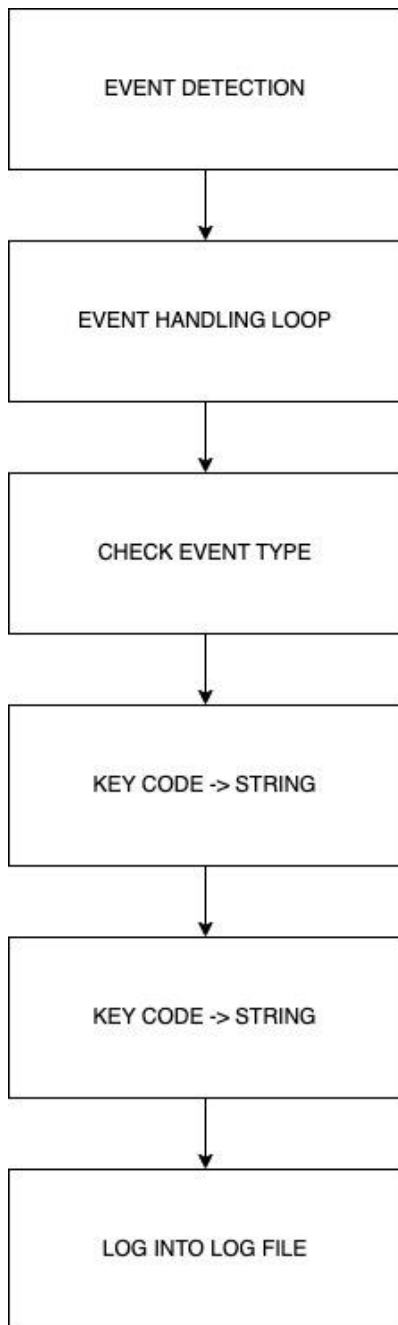


Fig2. After Event Detection

Pseudocode

1. Include necessary header files.
2. Define the main function:
 - a. Set up the event mask to capture keydown events and flags changes.
 - b. Create a CGEventTap to monitor keyboard events.
 - c. Check if the event tap creation was successful.
 - d. Create a run loop source and add it to the current run loop.

- e. Enable the event tap.
 - f. Check if a command-line argument is provided:
 - i. If "clear" is provided, clear the log file.
 - ii. Otherwise, set the log file location.
 - g. Get the current time and open the log file in append mode.
 - h. Check if the log file was opened successfully.
 - i. Write a header in the log file indicating that keystrokes are being recorded.
 - j. Print the log file location to stdout.
 - k. Run the CFRunLoop.
3. Define the CGEventCallback function:
 - a. Check if the event type is one of interest (keydown, flags changed, keyup).
 - b. Get the key code from the event.
 - c. Convert the key code to a string representation.
 - d. Write the string representation to the log file.
 - e. Return the event.
 4. Define the convertKeyCode function:
 - a. Match the key code to its corresponding character or symbol.
 - b. Return the string representation of the key code.
 - c. If the key code does not match any known value, return "[unknown]".

IV. RESULTS AND DISCUSSION

The keylogger software described in the provided code snippet functions by intercepting keyboard events on macOS systems and logging the keystrokes into a designated log file. Upon execution, the software initializes an event tap using the Core Graphics framework, which allows it to monitor low-level system events such as key presses and modifier key changes. As the user interacts with the keyboard, the event tap captures these events, triggering a callback function that processes the intercepted keystrokes.

Once intercepted, the keystrokes are processed by the event callback function, which extracts the key code associated with each pressed key using Core Graphics functions. Subsequently, the extracted key codes are converted into human-readable characters using a mapping function. This conversion enables the keylogger to translate the raw key codes into recognizable alphanumeric characters and special symbols, making the logged data more comprehensible.

The processed keystrokes are then logged into a specified log file using standard file I/O operations. Each keystroke is appended to the log file along with a timestamp indicating the time of occurrence. This logging mechanism

[illegible]

Comparative analysis of the keylogger software reveals several noteworthy aspects. Firstly, the software leverages the Core Graphics framework to intercept keyboard events, demonstrating a reliance on macOS-specific APIs for system-level event monitoring. This approach ensures compatibility with macOS systems and facilitates seamless integration with the operating environment.

V. CONCLUSION

As highlighted in the abstract, keyloggers discreetly capture keystrokes, enabling the monitoring of user actions without their explicit knowledge. This capability proves instrumental in recovering data compromised due to various factors, including power loss, by logging keystrokes and facilitating the retrieval of crucial information such as emails or URLs.

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